

Abhishek Gupta

AI Engineer

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Professional Summary

AI Engineer with hands-on experience designing and shipping production-grade agentic AI systems, MCP, RAG pipelines, and LLM-powered applications, built on 2+ years of professional backend engineering (Java, Spring Boot, AI Stack). Over the past 10 months, built and deployed multi-agent LangGraph systems with hybrid retrieval, dual-temperature inference, and full CI/CD to AWS - combining backend engineering discipline with applied ML, NLP, and Transformer expertise.

Skills

- **Programming Languages/Databases:** Java, Python, MySQL, SAP HANA, MongoDB, PostgreSQL
- **Libraries:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Seaborn, Hugging Face, Transformers
- **AI Security:** LangSmith (Tracing, Evaluation), AWS Bedrock Guardrails, Guardrail AI, Model Versioning, Pipeline Monitoring
- **AI Stack/Frameworks:** LangChain, LangGraph (Multi-Agent Orchestration), RAG, Prompt Engineering, Fine-Tuning, Model Context Protocol, FastMCP
- **Cloud & Deployment:** Git & GitHub, Docker, GitHub Actions, Docker Compose, AWS (EC2, S3, AWS Bedrock)
- **Vector Databases:** ChromaDB, FAISS, Qdrant, Pinecone, Neo4j
- **AI Development Tools:** GitHub Copilot, Cursor, Claude Code, Codex
- **Hard Skills:** Gen AI System Design, Debugging, Agile Methodology, Data Structures, Event-Driven Architectures, Technical Documentation

Experience

Software Engineer

Asint Pvt.Ltd – Ahmedabad, Gujarat | Aug 2024 – Present

- Designed and built risk-based inspection APIs for an analytics-driven reliability platform, **applying rule-based scoring to reduce operational risk by 60%**.
- Implemented the retrieval pipeline for an agentic Graph RAG-based assistant, integrating Neo4j graph database queries with LangChain/LangGraph for context retrieval.
- Worked on parts of the multi-agent orchestration layer for the same assistant, contributing to agent coordination and task routing logic.
- Optimized backend APIs, improving REST performance through query tuning and **response efficiency by 80%**.
- **Owned end-to-end backend development across multiple production systems, translating business rules into reliable, scalable RESTful APIs in Java and Spring Boot.**
- **Debugged & Resolved complex production issues through** root-cause analysis, and performance tuning.

AI-Powered Projects

EquityLens — Agentic Market & Company Research Analyst | July 17 – August 3, 2026

Tech Stack: Claude (Anthropic API) · RAG · Agentic Workflow · MCP · FastAPI · ReactJs · Qdrant / Pg Vector · LangSmith · AWS

- **End-to-end agentic research assistant:** Designed and built a full-stack agentic research analyst, pairing a FastAPI backend with a React frontend, that takes a market or company research query and autonomously plans, gathers, cross-verifies, and synthesizes findings from SEC filings, live market data, and news sources into a fact-checked research output.
- **5-agent LangGraph orchestration pipeline:** Architected a 5-agent LangGraph pipeline — a planner agent that decomposes the research query, three parallel specialist agents covering SEC filings, market data, and news, a synthesizer agent that merges their outputs, and a fact-checker agent — connected with conditional routing so the graph adapts its path based on each agent's output.
- **Bounded self-correction loop for factual accuracy:** Implemented a bounded correction loop between the synthesizer and fact-checker agents: claims in the synthesized output are verified against source data, and any unverified or contradicted claim is sent back for revision within a capped number of retries, preventing both unchecked hallucination and infinite looping.
- **Hybrid RAG over SEC filings:** Designed hybrid retrieval-augmented generation over SEC filings using Qdrant / Pg Vector, combining dense semantic embeddings with keyword-based retrieval to surface precise, citation-grounded passages from lengthy regulatory filings for the agents to reason over.
- **Full-stack observability with LangSmith (Evaluation harness):** Instrumented full observability via LangSmith, capturing per-node, per-LLM-call, and per-tool-call traces, including latency, token usage, and cost at each step of the pipeline — giving granular visibility into where time and spend go across a multi-agent run.
- **MCP-based live data integration:** Integrated pre-built MCP (Model Context Protocol) tool servers to give agents direct, structured access to live financial data and breaking news retrieval, extending the system's knowledge beyond static filings to real-time market context via Claude's tool-use API.

Synthr — Agentic SWE AI Assistant | May 15 – May 31, 2026

Tech Stack: Python | FastAPI | LangGraph | Anthropic Claude | Qdrant | React | PostgreSQL | Docker | AWS | LangSmith.

- **End-to-end autonomous SWE assistant:** Designed and built a full-stack AI assistant, pairing a responsive React frontend with an asynchronous, event-driven FastAPI backend, that autonomously resolves GitHub tickets end-to-end — ingesting an issue, classifying developer intent, retrieving relevant codebase context via hybrid RAG, generating and editing code, validating changes with automated test runs inside Docker containers, and opening a pull request via PyGitHub, without human intervention at any step.
- **6-agent LangGraph orchestration pipeline:** Architected a 6-agent LangGraph pipeline (intent classification, context retrieval, planning, code generation, test execution, review/retry, and PR authoring) with conditional routing between agents, automated retry loops on test or validation failure, and structured Pydantic outputs enforced via Claude's with_structured_output API across every agent — ensuring type-safe, reliable hand-offs through the whole workflow
- **Hybrid RAG for codebase retrieval:** Implemented hybrid retrieval-augmented generation using Qdrant, combining dense embeddings with sparse BM25 vectors and fusing results via Reciprocal Rank Fusion (RRF), followed by Cohere reranking for precision. Used Tree-sitter AST-based chunking to split source files along syntactic boundaries (functions/classes) rather than arbitrary token windows, significantly improving retrieved-context relevance for code.

- **Continuous repository synchronization:** Built an incremental re-embedding pipeline that keeps the vector index in sync with the live codebase: GitHub webhooks fire on every push, and asynchronous Celery workers pick up the diff, re-chunk and re-embed only the changed files, and update Qdrant — avoiding full repo re-indexing while keeping retrieval current.
- **Observability & production deployment:** Integrated LangSmith for end-to-end tracing and observability across all agents in the pipeline, and shipped the system to production with multi-stage Docker builds, Docker Compose orchestration, and GitHub Actions CI/CD, deployed on AWS EC2 with PostgreSQL (Alembic migrations) and Redis backing the application, and Docker Hub for container image distribution.

Nexus AI | Apr 12 – Apr 24, 2026

Tech stack used: Python | FastAPI | Claude API | Claude Vision | ChromaDB | MiniLM | PostgreSQL | Docker | SSE

- Built a full-stack **Multimodal AI Assistant** with **RAG** document analysis, image understanding via Claude Vision, and real-time token streaming via **Server-Sent Events**.
- Implemented a complete **RAG pipeline**: PDF ingestion with PyMuPDF → semantic chunking → MiniLM embeddings → ChromaDB vector store → Claude context injection with page-level citations.
- Integrated Claude Vision API for multimodal image QA; designed scoped retrieval filtering by user_id and file_id for per-document accuracy.

Education

Lovely Professional University - 2026 Master Of Computer Applications (Data Science, Online) - 2026 Pursuing

Indira Gandhi National Open University - 2025 Bachelor Of Business Administration - 61%